MILITARY SPECIFICATION
FOR
CASE, CARTRIDGE, 37MM, MK1A2

1. SCOPE

1.1 This specification covers Case, Cartridge, 37MM, MK1A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issure In effect on date of invitations for bids form a part Or this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-G-2550	-	General Specification for Ammunition, Except Small Arms Ammunition.
MSL-P-10025	-	Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Inspection Terms and Definitions
MIL-STD-639	-	Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Ordnance

F-8596159	Case, Cartridge, 37MM, MK1A2.
F-7548235	Cartons, Ammunition, for Interplant Shipment of Cartridge Cases, Empty - 37MM to 120MM Inclusive-.
IEL-F-8596159	Inspection Equipment List for Case, Cartridge, 37MM, MK1A2.

PUBLICATIONS

ORD-M608-11	Ordnance Inspection Handbook - Procedures and Tables for Continuous Sampling by Attributes.

FSC 1310

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General. - The cartridge case shall comply with the requirements of the applicable drawings, referenced specifications and the follwing:

3.2 Preproduction Sample. - Unless otherwise directed by the contracting officer, a preproduction sample shall be required.

3.3 Material. - The contractor shall be responsile for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Annealing. - Annealing temperature shall not exceed 1200 degrees Fahrenheit(F).

3.5 Stress Relief. - After the last cold forming operation, the cartridge case shall be given a stress relief treatment of approximately 500 degrees Fahrenheit (F) for one hour.

3.6 Mouth Anneal. - After the final forming operation, the cartridge case shall be annealed at the mouth for a distance of 0.6 inches minimum, 1.0 inches maximum

.3.7 Mercurous Nitrate. - The cartridge case, when fitted with a steel mouth plug simulating the 37MM, TP, M63, Mod. I, Projectile, maximum dimensions, shall show no evidence of cracking after immersion in mercurous nitrate.

3.8 Maximum Profile. - The cartridge case shall freely enter the Ordnance certified maximum profile gage in such a manner as to permit the slide to close without interference.

3.9 Mechanical Propertied.

3.9.1 Hardness. - The metal,l/16 inch minimum, from the surface around the primer hole shall have **a** Rockwell hardness value of not less than B65.

3.9.2 Tensile Strength. - Specimens taken from sample cartridge cases in accordance with Figure I shall comply with the requirements of Table 1, (see 6.1).

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TABLE I

	Pounds Per Square Inch (PSI)
Specimen A	44,000 Minimum - 54,000 Maximum
Specimen B	47,000 Minimum -
Specimen C	62,000 Minimum -
Specimen D	62,000 Minimum -

Elongation shall be recorded for information only.

3.10 Functional. - The cartridge case shall be subjected to a functional test in the following phases:

a. Excess Pressure Phase
b. Excess, Excess Pressure *Phase* (for information only)

3.10.1 Obturation. - The cartridge case shall effectively seal the breech of the weapon against leakage of hot gases generated during firing.

3.10.2 Soundness. - After firing the cartridge case shall not contain splits or cracks, in the sidewall or head.

3.10.3 Extraction. - When fired, the cartridge shall be extracted by the service extractors when the weapon is operated in the normal manner (see 6.2).

3.10.4 Primer Retention. - The cartridge case shall retain the primer during firing and extraction.

3.11 Workmanship - - The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal Defects. - The cartridge case shall be free from cracks, blisters, seam, laminations, deep scratches and porosity.

3.11.1.1 Folds and Cold shuts. - The cartridge case shall contain no folds or cold shuts in the radius area extending beyond the tangents from the base and sidewall, however, at no time will anu fold or cold shut exceed .010 maximum length, See Figure II.

3.11.2 Surface Finish. - The requirement for surface finish are as detailed on the applicable drawing.

3.11.3 Burr. - No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.11.4 Foreign Matter. - No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

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4. **QUALITY ASSURANCE PROVISIONS**

4.1 General Quality Assurance Provisions. - The quality assurance provisions in this section and other documents referenced in this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 to define the inspecion terms used. Inspection shall be in accordance with specification MIL-G-2550 except as specified herein.

4.1.1 Contractor Inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier my utilize his own facilities or any commercial laboratory acceptable to the Government Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor Quality Assurance System. - The contractor shall provide and maintain an adequate quality assurance system, acceptable to the Government, covering the supplies under the contract A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by this specification and other applicable documents referenced in this specification. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.1.3 Government Verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government product Inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed upon procedures or instances of poor practices which might have a detrimental effect upon quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Preproduction Inspection. - Prior to commencing regular production, the contactor shall submit a preproduction sample to a designated Government approved facility for evaluation. The sample shall consist of seventy (70) cartridge cases. The sample shall be manufactured in the same manner, using the same materials, equipment, processing and procedures as will be used in

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regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and previsional acceptance at source, the preproduction sample shall be inspected for all requirements of the drawings and specifications including all those characteristics listed in 4.3.3 and 4.3.4. The Government reserves the right to require new pre-production samples until such time as an acceptable sample is submitted. Sixty of the above cartridge cases shall be forward to a designated proving ground for ballistic testing.

4.3 Inspection Provisions. - Alternative inspection procedures and inspection equipment may be used by the contractor when such procedures and equipment provide as a minimum, quality assurance required in the contractual documents. The inspection procedures or equipment specified in this specification will prevail in case of dispute.

4.3.1 Material. - Reports of objective tests performed on the material by the contractor shall be required to insure compliance with the chemical and physical requirements. Reports shall include positive identification of the material and part, the applicable requirements, the specified tests performed and the results obtained. Numerical results of each specimen or sample inspected shall be included except that where numerical results are impossible an adjective description shall be used. Initial test reports shall accompany the preproduction sample. Prior to inspection, the inspector shall determine from the information stated on the test reports whether the material submitted complies with the requirements. Components made from material which does not comply shall be rejected.

4.3.2 Lot.

4.3.2.1 Submittal of Product. - The contractor shall submit the product in accordance with Standard MIL-STD-105 or ORD-M608-11 as applicable.

4.3.2.2 Size of Lot. - For the purpose of identification and shipment, the lot size shall be as agreed upon between the contractor and the contracting officer, provided the homogeneity of the inspection lot has been retained

4.3.3 Examination - Examination for major and minor defects shall be preformed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or ORD-M608-11 as applicable. an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance standard finish blocks prescribed in Standard MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

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4.3.3.1 Maximum Profile Gage. - Each cartridge case shall be, gaged in an Ordnance certified maximum profile gage. Failure of any cartridge case to comply with the requirements of 3.8 shall be cause for rejection of the cartridge case.

4.3.3.2 Classification of Defects.

4.3.3.2.1 Case, Cartridge, Dwg. F-8596159.

Defect		Method of Inspection
Major		
1.	Primer hole not functionally concentric with body	Gage
2.	Primer hole counterbore not functionally concentric with primer hole	Gage
3.	Diameter of Mouth	Gage
4.	Diameter forward of flange, min.	Gage
5.	Diameter of flange, min.	Gage
6.	Diameter of primer hole	Gage
7.	Diameter of primer hole counterbore	Gage
8.	Taper of primer hole	Gage
9.	Depth of primer hole counterbore, min.	Gage
10.	Thickness of wall at mouth	Gage
11.	Thickness of wall at rear datum	Gage
12.	Thickness through head, min.	Gage
13.	Thickness of flange, min.	Gage
14.	Length, Total	Gage
15.	Contour of inside of head and sidewall not smooth	Visual
16.	Metal Defective	Visual

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Minor

41.	Diameter at center datum length	Gage
42.	Diameter at rear datum length	Gage
43.	Length to forward datum diameter	Gage
44.	Width of shoulder in primer hole counterbore, min.	Gage
45.	Depth of primer hole counterbore, max.	Gage
46.	Angle at head, missing or incorrect	Visual
47.	Marking missing, incorrect or illegible	Visual
48.	Radii or chamfers missing or incorrect	Visual
49.	Surface finish, improper	Visual
50.	Burr	Visual
51.	Foreign matter	Visual

4.3.3.2.2 Packed Carton, prior to Sealing, Dwg. F-7548235

Defect	Method of Inspection

Major

None

Minor

41.	Support missing or damaged	Visual
42.	Separator missing or damaged	Visual
43.	Improper assembly	Visual
44.	Incorrect number in carton	Visual

4.3.3.2.3 Packed Carton, after Sealing, Dwg. F-7548235

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Defect	Method of Inspection
Major	
None	
Minor	
41. Carton damaged, cut or gouged through all layers	Visual
42. Carton not completely sealed	Visual
43. Contents loose	Manual
44. Marking missing, incorrect or illegible	Visual

4.3.4 Inspection Testing. - The following tests shall be performed in accordance with the provisions of 4.4, except as provided in 4.3.

4.3.4.1 Mercurous Nitrate - Five cartridge cases from the preproduction sample and five cases from each production lot shall be subjected to mercurous nitrate immersion. Failure of one cartridge case to comply with the requirements of 3.7 shall be cause for rejection of the sample or lot.

4.3.4.2 Mechanical Properties.

4.3.4.2.1 Hardness. - Five cartridge cases from the preproduction sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one cartridge case to comply with the requirements of 3.9.1 shall be cause for rejection of the sample or lot.

4.3.4.2.2 Tensile Strength. - Five cartridge cases from the Preproduction sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one cartridge case to comply with the requirements of 3.9.2 shall be cause for rejection of the sample or lot.

4.3.4.2.3 Elongation. - The percentage of elongation of each specimen used for tensile strength test shall be measured and recorded for information only.

4.3.4.2.4 Macro Examination - Five cartridge cases from the preproduction sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one cartridge case to comply with the requirements of 3.11.1.1 shall be cause for rejection of the sample or lot.

4.3.4.3 Functional. - The following functional tests will be performed by the Government.

4.3.4.3.1 Preproduction Sample. - The quantities listed in Table II will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criterial set forth in Table III except that no retest will be permitted.

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TABLE II

Test Phase	Quantity to be Tested
Excess Pressure Phase	30
Excess, Excess Pressure Phase (for information only)	30

4.3.4.3.2 Production Lot Sample. - Thirty cartridge cases from each production lot of 25,000 or less will be ballistically tested in the excess pressure phase. Lots which exceed 25,000 cartridge cases, one additional cartridge case will be tested for every one thousand cases (or fraction thereof) above 25,000. Acceptance and rejection criteria shall be in accordance with Table III.

TABLE III

Defect Characteristic	Accept	Retest	Cumulative Rejects
1. Obturation	0	1	2
2. Splits or cracks within 1 caliber of head	0	0	1
3. Splits or cracks in sidewall more than 1 caliber from head	1	2	3
4. Extraction	0	1	2
5. Primer retention	0	1	2

4.3.5 Process Control. - Monitoring at unscheduled intervals will be conducted for the following process operations to determine compliance with following requirements.

Annealing - 3.4

Stress Relief - 3.5

Mouth Anneal - 3.6

Failure to comply with the requirements shall be cause for rejection of all units processed since the last monitored approval of the process.

4.3.6 Inspection Equipment - The examination and tests shall be made using equipment listed on IEL-8596159, except as provided in 4.3. The

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Government will determine by means of its monitoring program, that the contractor has available and correctly utilizes gaging, measuring and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of. the required accuracy. The contractor shall have available a set of master gages, standards and other appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and be made available to the Government for review. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.4 Test Methods and Procedures.

4.4.1 Mercurous Nitrate. - The cartridge case fitted with a steel mouth plug in accordance with Figure III, shall be pickled in a forty percent solution (specific gravity 1.25) of commercial nitric acid for 30 seconds, and then rinsed in running water. The cartridge case shall then be immersed in a one percent solution of mercurous nitrate acidulated with one percent of nitric acid (specific gravity 1.42) for a period of 30 minutes. After the required immersion, the metallic mercury shall be volatized from the cartridge case on a hot plate and the case examined for cracks, (see 6.4).

4.4.2 Hardness. - The cartridge case shall be sectionalized through the center of the head. The hardness readings shall be taken on the cross sectional area through the primer boss. Hardness readings shall be taken on both sides within 1/16 inch from primer hole, in accordance with procedures for these tests prescribed by Federal Standard No. 151.

4.4.3 Tensile Strength and Elongation. - The test specimens shall be taken from the sample cartridge cases in accordance with the standard positions and size of tensile specimen chart Figure I. The tensile and elongation shall be determined in accordance with the procedures for these tests prescribed by Federal Standard No. 151.

4.4.4 Macro-Examination. - The cartridge case shall be sectionalized through the center of the head and polished, then examined for folds and cold shuts in the radius area of the head. One hundred magnification (100X) or more shall be used for this examination.

4.4.5 Functional.

4.4.5.1 Reproduction Sample.

4.4.5.1.1 Excess Pressure Phase. - The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

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4.4.5.1.2 Excess, Excess Pressure Phase. - The cartridge case will be assembled as required for service use. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4,4.5.2 Production Sample

4.4.5.2.1 Excess Pressure Phase. - The cartridges case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a servicable weapon for which the round is standard. The weapon will be operated in the service manner.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A- Not applicable.

5.1.2 Level B- Not applicable.

5.1.3 Level C - Cartridge cases shall be packed in accordance with requirements of drawing F-7548235.

5.2 Labeling and Marking. - Labeling and marking for shipment shall be in accordance with MIL-P-10025.

5.3 Data Card. - Data card information shall be as specified in specification MIL-G-2550.

6. NOTES

6.1 It is not implied that cartridge cases that contain minimum tensile strength values for all specimen. will function satisfactorily in ballistic test.

6.2 The term "extraction" shall mean that the fired cartridge case is loosened and palled to the rear of the weapon sufficiently to permit removal by hand.

6.3 The term "ejected" shall mean that the cartridge case is thrown free of the breech of the weapon to permit loading of the subsequent round.

6.4 Mercurous Nitrate - The pickling of cases in 40 percent nitric acid must be done under laboratory conditions by experienced technicians. Proper ventilation is assential, including suitable apparatus for the work. The removal of metallic mercury from the case must be done in a well ventilated area under a hood or exhaust fan. Technicians should avoid breathing in these vapors.

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Custodian:

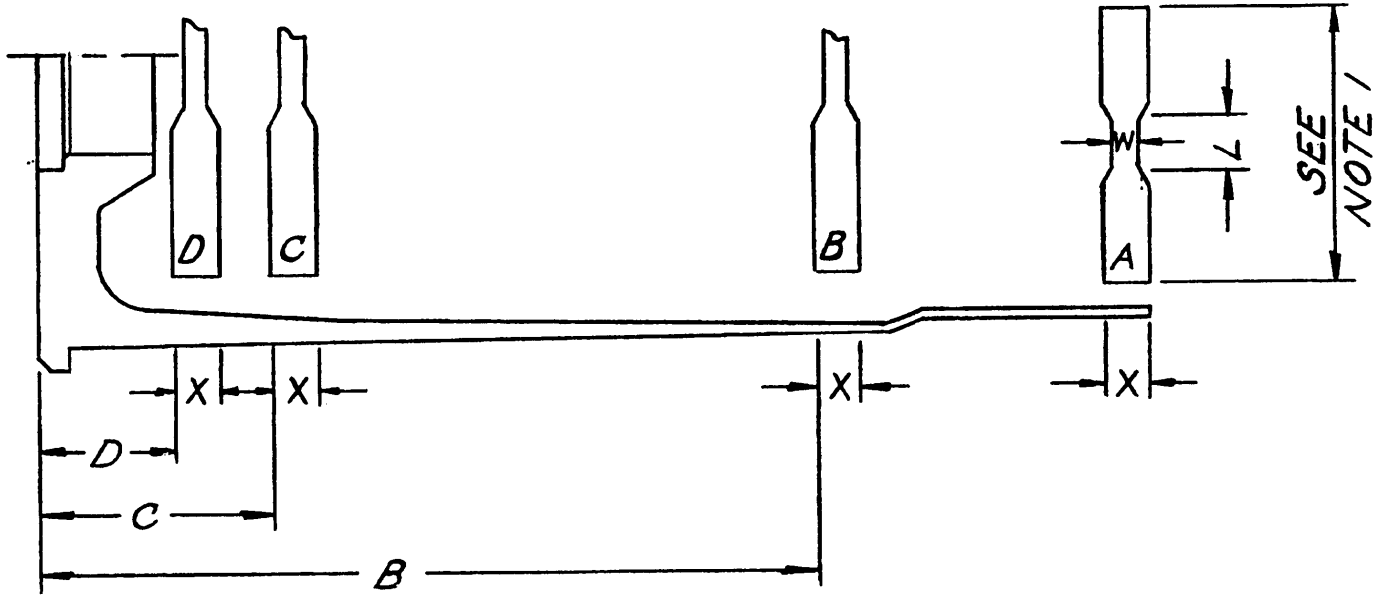
Army Ordnance Corps

Preparing Activity:

Army Ordnance Corps

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STANDARD POSITIONS AND SIZES OF TENSILE SPECIMENS FOR CASE, CARTRIDGE, 37MM, MK1A2



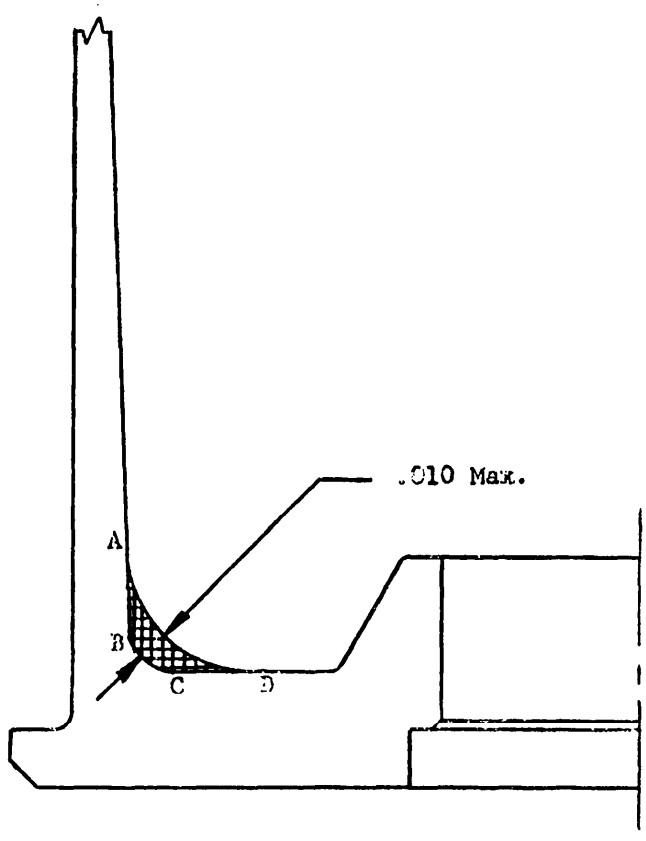
SPECIMEN	X	W	L	LOCATION FROM BASE
A	0.312	0.2	1.0	At Mouth
B	0.312	0.2	1.0	3.00
C	0.312	0.2	1.0	1.25
D	0.312	0.2	1.0	0.275

NOTE:

1. Overall length of specimen to suit fixture in testing machine.
2. Specimens C and D to be machined to uniform thickness.

FIGURE I

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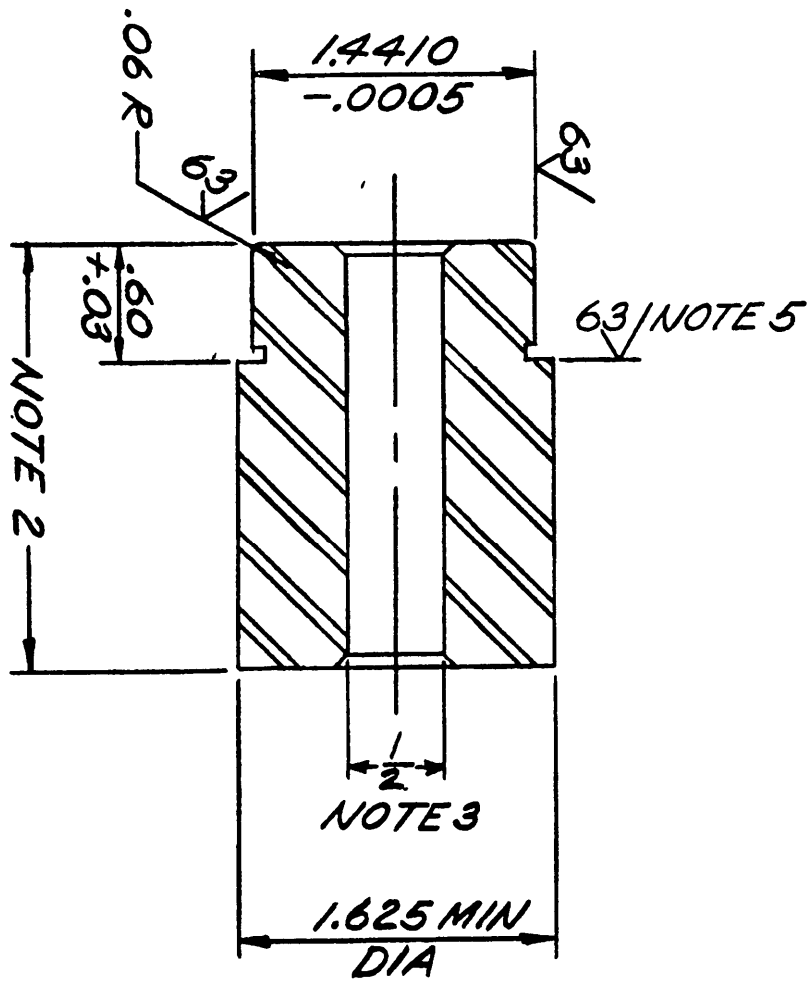


MACRO EXAMINATION

Folds or Cold Shuts shall not extend beyond shaded area inclosed by tangent AB, Tangent CD and Arc BC.

FIGURE II

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NOTES:

1 - Material:- Steel
2 - Overall length of specimen to suit testing procedure
3 - Centers permitted; hole may be threaded, for extracting plug
4 - 250/all over except as noted
5 - Cartridge case to be seated against this surface

MOUTH PLUG FOR USE IN MERCUROUS NITRATE TEST

FIGURE III

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